User Manual

AGRIMATE®

Maximum **Power** - Super **Economy**

AM-708T

Knapsack Motorised Sprayer



IS:3906

CM/L 8548088

DISCLAIMER: DUE TO CONSTANT UPGRADATION, FEATURES AND SPECIFICATIONS ARE SUBJECT TO CHANGE WITHOUT NOTICE

BEFORE OPERATING THIS MACHINE,
PLEASE READ THESE INSTRUCTIONS CAREFULLY

SAFETY PRECAUTIONS

Warning labels bearing the mark on the product and in this Manual refer to important safety matters. Please familiarize your self with them and observe them well.

If a warning label on the product becomes soiled or detached, please order a replacement label from your dealer and attach it in the requisite place

PRECAUTIONS

In this Operator's Manual, precautions that are considered to be particularly important are indicated as follows

WARNINGS IN THE MANUAL

⚠ WARNING

Failure to observe the precaution may result in the risk of death or serious injury.

❗ IMPORTANT

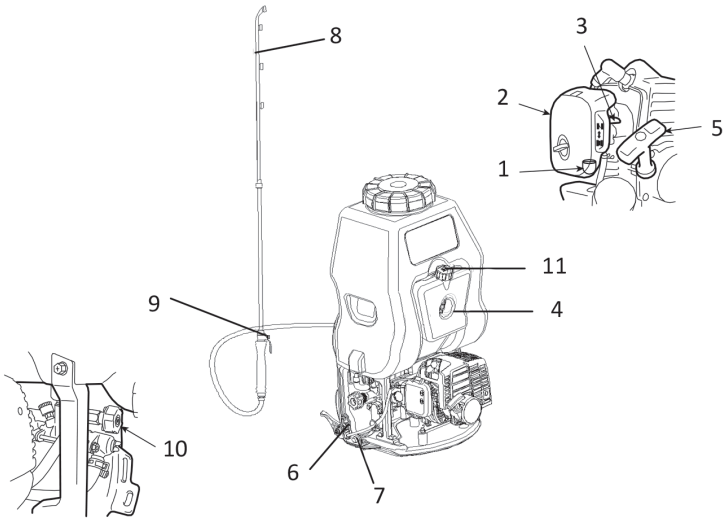
Failure to observe the precaution may result in damage or mal function of the product.

NOTE

Other additional explanations that may be of use when operating the product.

PARTS LOCATION

(1) Primer
(2) Air cleaner cove
(3) Choke lever
(4) Fuel tank
(5) Starter knob
(6) Throttle lever
(7) Stop switch
(8) Spray rod
(9) Grip
(10) Pressure regula
(11) Fluid drain cap



SPECIFICATION

Model name & number		**AM-708T Knapsack Motorised Sprayer**
Purpose		Spraying agricultural chemicals and cold water for pest control, weeding, and growth control
Dry weight * (kg)		9.8
External dimensions of main unit (length x width x height)** (mm)		335x390x640
Capacity of chemical fluid tank (litres)		25
Capacity of fuel tank (litres)		0.9
Pump	Type	Plunger type
	Speed (rpm)	1739
	Spray output Normal (MPa)	0.5 - 3.5
	Maximum (MPa)	3.5
	Maximum water absorption (litres/ min)	47.0
	Motor transmission system	Centrifugal clutch
Engine	Type	Single-cylinder air-cooled 2-cycle gasoline engine
	Engine number	AM262
	Exhaust capacity (cc)	25.6
	Fuel used	Lubricating oil/gasoline mix
	Lubricating oil used	2-cycle engine oil
	Admixture ratio	Gasoline 50 : Oil 1 (when using FC grade genuine oil)
	Carburetor	Diaphragm
	Ignition system	CDI
	Spark plugs	CMR7H
	Starter system	Recoil starter type
	Stopping system	Primary cable short circuit
	Air cleaner	Dry type

* When nozzle (supplied) is attached.

** Nozzle and back belt are not included.

Due to product improvements, these specifications are subject to change without notice.

SYMBOLS IN THE MANUAL



(1) (2) (3) (4) (5)

(1) Read owner's manual before operating this machine.
(2) Wear head, eye and ear protection.
(3) Wear foot protection.
(4) Wear gloves.
(5) Warning / Attention

Never modify your machine. We won't warrant the machine, if you use the remodeled power sprayer or if you don't observe the proper usage written in the manual

WARNINGS IN THE MANUAL

⚠ WARNING

Failure to observe the precaution may result in the risk of death or serious injury.

IMPORTANT

Failure to observe the precaution may result in damage or mal function of the product.

NOTE

Other additional explanations that may be of use when operating the product.

SYMBOLS ON THE MACHINE

For safe operation and maintenance, symbols are carved in relief on the machine. According to these indications, please be careful not to make a mistake.

The port to refuel the "MIX GASOLINE"
Position: FUEL TANK CAP

The direction to close the choke
Position: AIR CLEANER COVER

The direction to open the choke
Position: AIR CLEANER COVER

FOR SAFE OPERATION

Before using this product, please read this Operator's Manual thoroughly to ensure correct operation. The fol-lowing are precautions that should be observed in order to use the product smoothly. Elsewhere in the Manual, the ⚠ warning mark is used to bring your attention to these precautions.



BEFORE USING THE PRODUCT

- This product has been designed mainly for the purpose of spraying pest control chemicals and weedkillers. Please do not use it for any other purpose, as this may cause unforeseen accidents.
- Since this product uses agricultural chemicals that require careful operation, misuse can be dangerous. Do not use this product when tired or otherwise in poor physical condition, or when you may be unable to judge correctly or operate accurately, such as after taking cold relief medicines or drinking alcohol. Also, children or persons who are unable to understand the contents of this Manual should under no circum stances be permitted to use the product.



- The engine exhaust gas contains toxic carbon monoxide. Please do not use the product indoors, in vinyl houses or tunnels, or in other poorly ventilated places.



- Do not use this product in the following cases.

1. When the product could fall over or in other situations when it is difficult to hold or operate.
2. After sunset or at other times when eyesight is poor and the safety of the spraying area is difficult to as certain.
3. In worsening weather (rain, fog, strong wind, lightning, etc.)



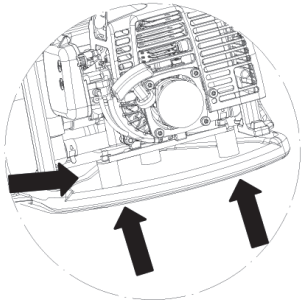
- When using for the first time, before commencing actual spraying please receive guidance in operating the product from a person with adequate experience.
- Fatigue reduces powers of concentration and in creases the risk of accidents. When planning work, please allow ample time, setting a limit of 30-40 minutes for single spraying sessions and breaks of 10-20 minutes in between. Also, do not spray for more than 2 hours in any one day.



- Please take good care of this Operator's Manual and refer to it for additional information from time to time.
- When selling or lending this product, it should be accompanied by this Operator's Manual.

⚠ WARNING

- Check to see if the shock-absorbing rubber mounthas become cracked or otherwise damaged. Note that failing to replace this rubber mount when it has become cracked or damaged may cause the engine to come loose from its frame during use, thus resulting in possible serious bodily injury.
- If cracked, be sure to replace without delay.





⚠ WARNING

1. If you don't observe the working time, or working manner, Repetitive Stress Injury (RSI) could occur. If you feel discomfort, redness and swelling of your fingers or any other part of your body, see a doctor before getting worse.
2. To avoid noise complaints, in general, operate product between 8 am. & 5 pm. on weekdays and 9 am. to 5 pm. on weekends.

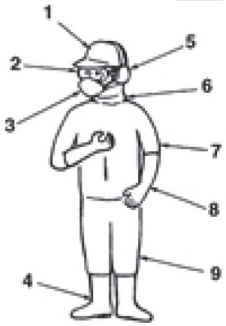
FOR SAFE OPERATION

❗ IMPORTANT

Check and follow the local regulations as to sound level and hours of operations for the product

PRECAUTIONS BEFORE SPRAYING

- Make sure you have the correct protective clothing, headgear, rubber gloves and boots, a protective mask, goggles, a towel, protective cream, etc., to prevent chemicals from coming into contact with your skin or entering your lungs. When spraying, please wear this protective gear to prevent any risk of injury.





1. Protective headgear
2. Protective goggles
3. Protective mask
4. Rubber boots
5. Ear muffs
6. Neck covering
7. Protective clothing
8. Rubber gloves
9. Protective clothing

- Please store chemicals in a locked storage box, locker, or other safe place out of the reach of children.
- Carefully read the instructions for using the chemicals, and familiarize yourself with the level of toxicity, method of use, etc.
- Please pay all due consideration to the area in which you will be spraying, to ensure that you do not pollute the water supply, rivers, lakes, ponds, etc., or cause any damage to residents, passers-by, livestock, etc.
- Prepare yourself physically on the day before spraying, by for example cutting down on alcoholic beverages or late-night work.
- Do not allow children, animals, etc., to enter the chemical spraying area.



- Before spraying, please prepare fully by inspecting sprayer nozzles and checking for abnormalities such as loose screws or spray hose connections, fuel leaks, defective or broken fluid gaskets, or damage to the carrying band.

PRECAUTIONS WHEN TRANSPORTING CHEMICALS

- When transporting chemicals, please handle with care to ensure that there is no spillage of chemicals from torn bags, broken bottles, or loose stoppers.
- Do not carry chemicals in the same container as food.

PRECAUTIONS WHEN TRANSPORTING THE SPRAYER

- Do not transport the sprayer while it has chemicals or fuel in the tank, as these could leak out as a result of impact during transportation.
- When transporting in a vehicle, please use a rope or similar to fix the sprayer firmly to the cargo trailer in an upright position. Please do not transport by bicycle or motorbike, as this is dangerous.

PRECAUTIONS CONCERNING FUEL

- The engine of this product is fueled by "mixed gasoline" that includes highly combustible fuel. Do not re-fuel or store fuel containers where there is any risk of combustion from incinerators, burners, bonfires, kitchen stoves, electric sparks, welding sparks, etc.



- Smoking while spraying or refueling is dangerous. Please avoid this at all costs.
- When refueling during use, always stop the engine and make sure there are no naked flames in the vicinity before refilling.
- Any fuel spilt during refueling should be wiped off completely with a rag or similar before starting the engine.
- After refueling, seal the fuel container, tightly close the fuel tank cap, and start the engine at least 3 meters away.

PRECAUTIONS WHEN REFILLING CHEMICALS

- Before filling the tank with chemicals, please make sure that the spray lever is completely at rest.
- After chemical refilling, firmly tighten the tank cap. If not fully tightened the cap may become loose during spraying, with the risk of physical exposure to the chemicals.

PRECAUTION WHEN STARTING THE ENGINE

- Please keep children, animals, etc. well away when starting the engine and spraying.

PRECAUTIONS WHEN SPRAYING

- Since protective clothing generally offers poor ventilation and thus places stress on the body, there is a risk of contracting heatstroke or other illness when operating for long periods under high temperatures. Please avoid spraying on hot days, aiming instead for early morning and late afternoon times when temperatures are relatively cool and winds are low.
- When starting work, please pay all due consideration to the spraying time, wind direction, etc., to ensure that no noise disturbance or chemical hazard is caused to nearby residents, passers-by, farm crops, and so on.
- While spraying please hold the grip firmly at all times and be careful not to point the nozzle in the direction of people or animals.
- While spraying please be careful of the wind direction and always stand on the upwind side, to avoid spraying yourself with chemicals.
- Should you begin to feel even slightly unwell while spraying, consult a doctor immediately. When doing so, inform the doctor of the name of the chemical you were using, the conditions of use, etc.
- If the sprayer is tilted the chemical may leak out through the air holes in the cap. When spraying always make sure you have a firm footing and maintain your balance.
- Do not touch the spark plugs or plug cords while the engine is running, as you may get an electric shock.



- Do not touch the muffler, spark plugs, or other metal parts with your bare hands while the engine is running or immediately after stopping the engine, as there is the risk of burns due to high temperatures.



PRECAUTIONS AFTER SPRAYING

- Please seal unused chemicals and store them out of the reach of children.
- Do not simply discard empty chemical containers, but dispose of them safely by incinerating, burying, or another similar method. In addition, please take every care that the water used to clean spraying tools does not create any hazard.
- Fully inspect and maintain protective clothing, masks, gloves, and other protective gear in preparation for the next session.
- After completing the work and tidying away chemicals and spraying tools, you should immediately take a bath or wash your hands, feet, face, etc. thoroughly with soap, as well as washing your mouth out.
- You should change all clothes worn, including underwear, and wash them thoroughly. Clothes worn during spraying should not be worn again the following day.
- After spraying, please limit your alcohol intake and rest thoroughly by going to bed early. If you feel even slightly unwell, please see a doctor as soon as possible.

OTHER PRECAUTIONS

- When engaged in spraying work over a protracted period of time, please have your health checked regularly.
- Please carry out spraying in a planned fashion, keeping a daily record of the date and time of spraying, the chemicals used, the target pest, contents of the work, duration of the work, and other details.
- Please keep the following points in mind when selecting or using protective gear.

a) Protective clothing
Choose clothing that is cool to wear and offers good ventilation and waterproofing. In addition, wear items that enable you to carry out spraying work in total comfort and safety.

b) Spraying hood
Use a hood with a brim that is waterproof and covers your neck and shoulders.

c) Gloves
Use gloves that are difficult for chemicals to penetrate and do not slip even if you perspire while doing the work.

d) Rubber apron
When mixing chemicals, wear a long, thick, and wide apron that cannot be penetrated by chemical splashes.

e) Protective goggles, mask, etc.
Select suitable protective goggles, mask, rubber boots, eye washer, and protective cream, etc., paying attention to standards and the recommendations of farm work safety promotion organizations.



MAINTENANCE PRECAUTIONS

- In order to maintain the functions of the product, regularly carry out the maintenance inspections listed in this Manual. When maintenance or parts replacement not indicated in this Manual are required, please consult your product supplier or nearest our product dealer.
- Never keep the engine running while carrying out in spections or maintenance.
- Do not modify the sprayer or dismantle the engine. This could lead to breakdown and serious accidents during operation.
- Do not touch the muffler or spark plugs with your bare hands immediately after stopping the engine. There is the risk of burns due to high temperature.



- For replacement parts, please use original parts or brands designated us

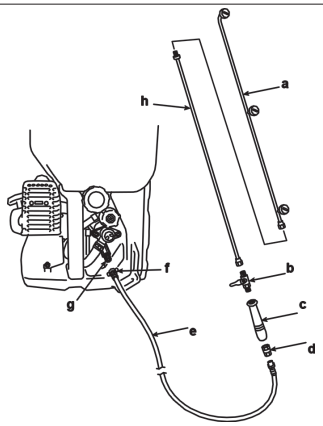
MAINTENANCE OF LABELS

1. Please keep warning labels clean and unmarked.
2. If warning labels become soiled or peel off, please order from your supplier and replace with new ones.
3. When affixing new labels please wipe off all dirt and moisture.

SET UP

- **Connecting the spray hose and nozzle**

a. spray rod
b. cock
c. grip
d. joint
e. spray hose
f. connecting nut
g. spray outlet
h. rod



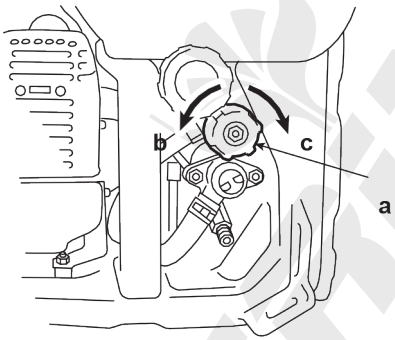
1. Connect the spray hose to the pump spray outlet and firmly tighten the connecting nut.
2. Tightly screw the joint onto the hose.
3. Tightly screw the grip onto the joint.
4. Tightly screw the cock onto the grip.
5. Tightly screw the rod onto the cock.
6. Tightly screw the spray rod onto the rod.
7. Connect the on-off switch and throttle lever to the body of the machine.

Adjusting spray pressure

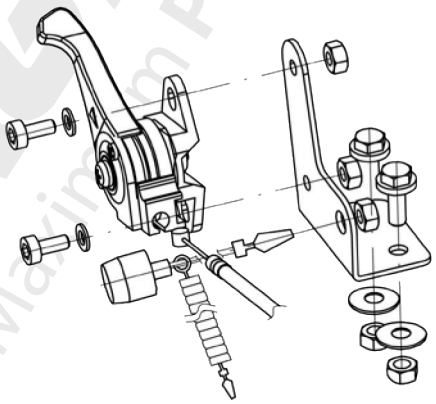
- this product has a pressure regulating dial to change spray pressure. Set the pressure at a level appropriate to the spray conditions.

Set the pressure regulating dial as follows:

- Turn in a clockwise direction to increase the pressure.
- Turn in a counterclockwise direction to decrease the pressure



a. Pressure regulating dial
b. Decreasing the pressure
c. Increasing the pressure



FUEL

⚠ WARNING

- Gasoline is very flammable. Avoid smoking or bringing any flame or sparks near fuel. Make sure to stop the engine and allow it cool before refueling the unit. Select outdoor bare ground for fueling and move at least 3 m (10 ft) away from the fueling point before starting the engine.



- The engines are lubricated by oil specially formulated for air-cooled 2-cycle gasoline engine use. Please use an anti-oxidant added quality oil expressly labeled for air-cooled 2-cycle engine use (JASO FC GRADE OIL or ISO EGC GRADE).
- Do not use BIA or TCW (2-stroke water-cooling type) mixed oil.



RECOMMENDED MIXING RATIO GASOLINE 50 : OIL 1 (when using genuine oil)

- Exhaust emission are controlled by the fundamental engine parameters and components (eq., carburation, ignition timing and port timing) without addition of any major hardware or the introduction of an inert material during combustion.
- These engines are certified to operate on unleaded gasoline.
- Make sure to use gasoline with a minimum octane number of 89RON (USA/Canada: 87AL).
- If you use a gasoline of a lower octane value than prescribed, there is a danger that the engine temperature may rise and an engine problem such as piston seizing may consequently occur.
- Unleaded gasoline is recommended to reduce the contamination of the air for the sake of your health and the environment.
- Poor quality gasolines or oils may damage sealing rings, fuel lines or fuel tank of the engine.

HOW TO MIX FUEL

⚠ WARNING

- Pay attention to agitation.

1. Measure out the quantities of gasoline and oil to be mixed.
2. Put some of the gasoline into a clean, approved fuel container.
3. Pour in all of the oil and agitate well.

4. Pour in the rest of gasoline and agitate again for at least one minute. As some oils may be difficult to agitate depending on oil ingredients, sufficient agitation is necessary for the engine to last long. Be careful that, if the agitation is insufficient, there is an increased danger of early piston seizing due to abnormally lean mixture.
5. Put a clear indication on the outside of the container to avoid mixing up with gasoline or other containers.
6. Indicate the contents on outside of container for easy identification.

FUELING THE UNIT

1. Untwist and remove the fuel cap. Rest the cap on a dustless place.
2. Put fuel into the fuel tank to 80% of the full capacity.
3. Fasten the fuel cap securely and wipe up any fuel spillage around the unit.

⚠ WARNING

1. Select bare ground for fueling.
2. Move at least 3 meters (10 feet) away from the fueling point before starting the engine.
3. Stop the engine before refueling the unit. At that time, be sure to sufficiently agitate the mixed gasoline in the container.

FOR YOUR ENGINE LIFE, AVOID:

1. FUEL WITH NO OIL (RAW GASOLINE) - It will cause severe damage to the internal engine parts very quickly.
2. GASOHOL - It can cause deterioration of rubber and/or plastic parts and disruption of engine lubrication.
3. OIL FOR 4-CYCLE ENGINE USE - It can cause spark plug fouling, exhaust port blocking, or piston ring sticking.
4. Mixed fuels which have been left unused for a period of one month or more may clog the carburetor and result in the engine failing to operate properly.
5. In the case of storing the product for a long period of time, clean the fuel tank after rendering it empty. Next, activate the engine and empty the carburetor of the composite fuel.
6. In the case of scrapping the used mixed oil container, scrap it only at an authorized repository site.

NOTE

- As for details of quality assurance, read the description in the section Limited Warranty carefully. Moreover, normal wear and change in product with no functional influence are not covered by the warranty. Also, be careful that, if the usage in the instruction manual is not observed as to the mixed gasoline, etc. described therein, it may not be covered by the warranty.

CHEMICAL FILLING

⚠ WARNING

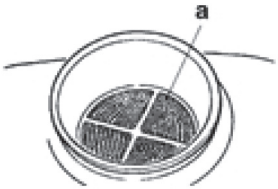
- Chemical substances should be handled correctly, observing the precautions marked on the container. Some chemicals are extremely hazardous to humans and animals, and handling errors can cause serious poisoning or even fatal accidents.
- When not in use, chemical substance containers should be stored separately from other containers out of the reach of children.

❗ IMPORTANT

- Do not fill the chemical tank with undiluted fluid. This could cause the gasket and valves to lose their durability

PROCEDURE FOR FILLING

1. Use a bucket or other suitable container and dilute the chemical to the requisite strength.
2. Place the sprayer on an even and stable surface and remove the chemical tank cap.
3. Fill the tank by passing chemical through the strainer.
4. When filling is complete, firmly tighten the cap.



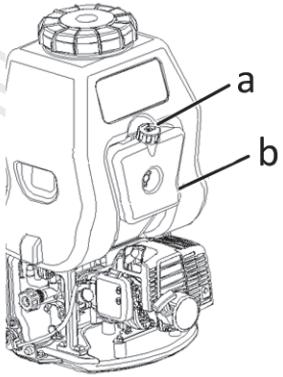
a. strainer

Chemical tank capacity: 25 litres

OPERATION

⚠ WARNING

- Highly combustible. When refueling, switch the engine off and keep well away from naked flames.
- After filling the tank with fuel, firmly tighten the cap.



a. cap
b. fuel tank

CHEMICAL FILLING

⚠ WARNING

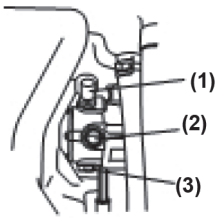
- Before starting the engine, make sure that the spray lever (spray valve) is completely returned. Starting with the spray lever still pulled out is dangerous, as chemical fluid may spray out of the nozzle as soon as the engine is started.



- To start the engine, place the sprayer on an even and stable surface or platform. If the sprayer is unstable, the reaction on starting etc. may cause it to fall.

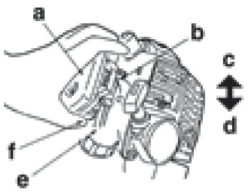
❗ IMPORTANT

- Do not allow the chemical fluid or water in the tank to run out while the engine is running. If the engine runs empty it may cause breakage of the spray pump.
- Make sure that the gear oil is filled until the middle of the check-window. If it is not enough, loose the upper bolt and fill the gear oil. (SAE40CC)



(1) bolt
(2) window
(3) drain bolt

1. Press the primer at the bottom of the carburetor a few times with your finger until the fuel starts to flow through the transparent pipe.

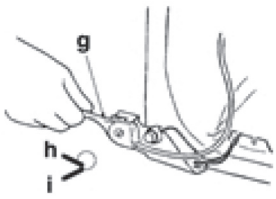


a. air cleaner
b. choke lever
c. close
d. open
e. transparent pipe
f. primer

NOTE

- There is no need to operate the primer when restarting immediately after stopping the engine, provided there is fuel left in the fuel tank.

2. Lift the choke lever on the right of the air cleaner until it is in the "closed" position, then open the throttle lever on the main unit side about 1/3-1/2 from the "slow" position.



g. throttle lever

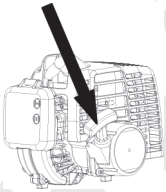
h. fast

i. slow

NOTE

- When restarting immediately after stopping the engine, set the choke lever to "open" and the throttle lever to "slow".

3. Place the sprayer in a stable location, and pull the starter knob with the right hand while holding the chemical tank down with the left hand. The starter knob should first be pulled out lightly, then, when resistance is felt, pulled more swiftly and vigorously



IMPORTANT

- Do not pull the rope out completely or release the knob so that the rope returns, as this can cause starter malfunction.

4. Once the engine has started, gradually open the choke, warm up at low speed for 1-2 minutes, then return the throttle lever to the "slow" position.

IMPORTANT

- If the rope is pulled continuously with the choke closed, the spark plugs may become flooded and the engine may fail to start. If so, you should first remove the spark plugs and dry the electrodes, then repeat the starting procedure.

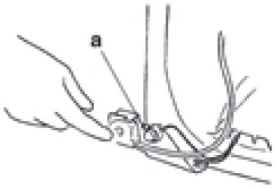
• Stopping engine

WARNING

- Stop the engine immediately in the event of an emergency.

1. Lower the throttle lever fully and cool the engine down for 1-2 minutes at low speed.

2. Hold down stop switch on the left side of the sprayer frame until the engine has come to a complete stop .



a. Stop switch

! IMPORTANT

- Except for an emergency, avoid stopping the engine while pulling the throttle trigger.

• Spraying

⚠ WARNING

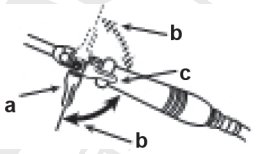
- Please use correctly, observing the precautions in this Manual ("For Safe Operation", p5 to 7).
- Before lifting the sprayer onto your back, check that the chemical tank and fuel tank caps are tightly closed.
- When spraying, do not point the nozzle at your face, or at other people, animals, etc.
- If the sprayer is tilted too much, chemical fluid or fuel may leak out through the air holes in the cap. When using, always try to maintain an upright posture.

! IMPORTANT

- If the chemical tank becomes empty while spraying, reduce the engine to low speed immediately. Empty running of the engine may cause the spray pump to break.
- Please avoid using multiple nozzles, jet nozzles, or other nozzles that give a large spray volume, as spraying efficiency is sometimes impaired owing to insufficient pressure.

• Spraying procedure

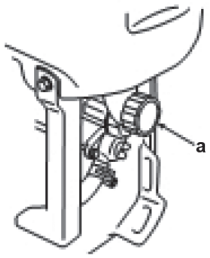
1. Start the engine, and lift the sprayer onto your back.
2. Adjust the carrying band to a suitable length for carrying.
3. Increase the engine speed, and open the spray cock. Chemical fluid starts to spray out from the nozzle in the form of mist.
4. When stopping work, close the spray cock fully and reduce the engine speed.



a. Spray cock
b. Close
c. Open

• Cleaning after completion of spraying

1. Loosen the fluid drain cap , transfer the chemical fluid left inside the tank into a suitable container, and process appropriately with all due consideration to safety.
2. Pour about 5 litres of clean water into the chemical tank, spray through the nozzle for 2-3 minutes, and drain the water left inside the tank.



a. Fluid drain cap

• Disposal

When disposing your machine, fuel or oil for the machine, be sure to allow your local regulations.

MAINTENANCE

	System/ components	Procedure	Before use	Every 25 hours after	Every 50 hours after	Every 100 hours after	Note
Engine	fuel leaks, fuel spillage	wipe out	✓				
	fuel tank, air filter, fuel filter	inspect/clean	✓	✓			replace, if necessary
	idle speed adjuster	see adjusting idling speed	✓				replace carburetor if necessary
	spark plug	clean and readjust plug gap			✓		GAP: 0.6 to 0.7 mm (.025 in) replace, if necessary
	cylinder fins, intake air cooling vent	clean		✓			
	muffler, cylinder exhaust port	clean				✓	
	throttle lever, stop switch	check operation	✓				
	screws/nuts/bolts	tighten/ replace	✓			✓	except adjusting screws
Pump	gear oil	replace				✓	

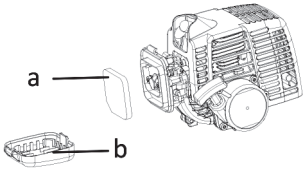
⚠ WARNING

- Make sure that the engine has stopped and is cool before performing any service to the machine. Contact with hot muffler may result in a personal injury.
- Do not modify the sprayer or dismantle the engine.
- When replacing parts please use our original parts or designated parts.

• Air cleaner

- After every 25 hours of operation, please remove the air cleaner cover and inspect the element. If it is too dirty, wash it carefully in warm water containing a neutral detergent, and return it to its original position after drying it thoroughly.
- If the element is distorted or damaged, please replace it with a new one.

a. Element
b. Air cleaner cover



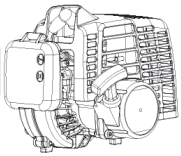
❗ IMPORTANT

- If the air cleaner element is blocked, the efficiency of the engine will be reduced. In addition, the engine interior will suffer abnormal wear if operated without the element or if continually operated with a distorted or damaged element fitted.

• Fuel filter

After every 25 hours of operation, empty the fuel tank, detach the fuel filter from the tank, and remove all dirt. If the filter is too clogged, please replace it with a new one.

a. Fuel filter



❗ IMPORTANT

- If the fuel filter is clogged, the engine speed may be limited or speed fluctuations may occur.
- If the engine is operated without a fuel filter, dirt will accumulate in the carburetor and cause it to malfunction.

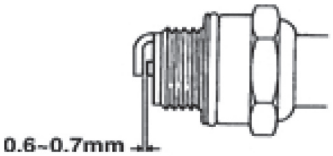
• Spark plug

⚠ WARNING

- Do not touch the spark plug with your bare hands immediately after operation, as there is the risk of burns due to high temperature.



- After every 50 hours of operation, detach the spark plug and remove dirt from the electrodes with a wire brush or similar.
- The correct electrode gap is 0.6 to 0.7mm.
- When replacing plugs, please use designated items.



IMPORTANT

- If too much fuel is absorbed or poor quality oil is used, the spark plug electrodes become dirty, making the engine harder to start.
- Note that using any spark plug other than those designated may result in the engine failing to operate properly or in the engine becoming overheated and damaged.
- To install the spark plug, first turn the plug until it is finger tight, then tighten it a quarter turn more with a socket wrench.

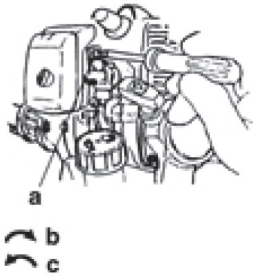
TIGHTENING TORQUE: 14.7 to 21.6 N.m. (150 to 220 kgf-cm)

• Adjusting the engine

Although the engine is adjusted on leaving the factory, after repeated use it may sometimes require readjustment or maintenance. Please consult your original supplier regarding inspections and maintenance other than those shown below.

- Idling speed

If the engine stalls or the pump continues to move when in idling position with the throttle lever completely lowered, readjust the engine speed using the idle speed adjuster on the left of the carburetor.



a. idle speed adjuster

b. increase speed

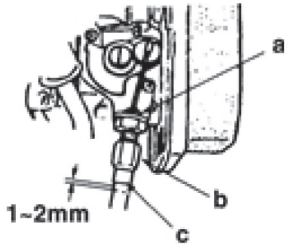
c. reduce speed

- Turn the idling adjustment screw:
  - counter-clockwise to reduce engine speed
  - clockwise to increase engine speed

NOTE

- Warm up the engine before adjusting the idling speed.
- Throttle wire play

Throttle wire play should be 1 to 2 mm. After completely returning the throttle lever, take the throttle wire at the carburetor end in your fingers and give a light tug. If the play is too great or too small, readjust the position of the wire retaining nut. Loosen the lock nut and turn the wire retaining nut clockwise (screw in) to increase the play and counter-clockwise (unscrew) to reduce it. After adjusting, please fasten the wire retaining nut by tightening the lock nut.



a. lock nut
b. wire retaining nut
c. wire sleeve

MUFFLER

WARNING

- Inspect periodically, the muffler for loose fasteners, any damage or corrosion. If any sign of exhaust leakage is found, stop using the machine and have it repaired immediately.
- Note that failing to do so may result in the engine catching on fire.

PROCEDURES TO BE PERFORMED AFTER EVERY 100 HOURS OF USE

1. Remove the muffler, insert a screwdriver into the vent, and wipe away any carbon buildup. Wipe away any carbon buildup on the muffler exhaust vent and cylinder exhaust port at the same time.
2. Tighten all screws, bolts, and fittings.
3. Check to see if any oil or grease has worked its way in between the clutch lining and drum, and if it has, wipe it away using oil-free, lead-free gasoline.



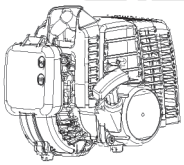
INTAKE AIR COOLING VENT

⚠ WARNING

- Never touch the cylinder, muffler, or spark plugs with your bare hands immediately after stopping the engine. The engine can become very hot when in operation, and doing so could result in severe burns.
- When checking the machine to make sure that it is okay before using it, check the area around the muffler and remove any wood chips or leaves which have attached themselves to the brushcutter. Failing to do so could cause the muffler to become overheated and that in turn could cause the engine to catch on fire. Always make sure that the muffler is clean and free of wood chips, leaves, and other waste before use.
- Check the intake air cooling vent and the area around the cylinder cooling fins after every 25 hours of use for blockage. Note that it is necessary to remove the upper cover and the lower cover in order to be able to view the upper part of the cylinder.

❗ IMPORTANT

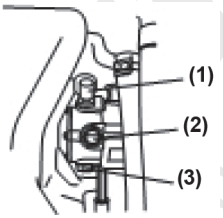
- If waste gets stuck and causes blockage around the intake air cooling vent or between the cylinder fins, it may cause the engine to overheat, and that in turn may cause mechanical failure on the part of the brushcutter.



(1) Intake air cooling vent

OIL

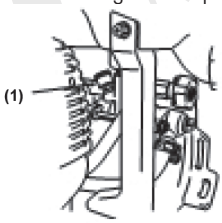
- After first 20 hours of use, replace the gear oil. After that, replace it after every 100 hours of use (SAE40CC, 100cc).



(1) bolt
(2) window
(3) drain bolt

GREASE

- Open the upper lid of the grease cup and check the grease amount. Fill it up if it is decreasing. (#3)



(1) grease cup

STORAGE

⚠ WARNING

- When draining fuel, please be careful not to spill it. If fuel is spilt, wipe off thoroughly. In addition, please be sure to close the storage container cap tightly.

❗ IMPORTANT

- If fuel is left in the engine for protracted periods, the inside of the carburetor may become clogged and cause engine malfunction (faulty starting and insufficient output).
- During storage, slightly loosen the cap of the chemical fluid tank. If screwed too tightly, the gasket may become distorted.

On completion of all work procedures, when not planning to use again for a protracted period please carry out the following pre-storage care, to prevent trouble caused by time-lapse changes.

1. Clean the chemical fluid tank and nozzle (see page 10).
2. Remove dirt from the sprayer, inspect it for any damage or looseness, and, if any abnormality is found, correct it thoroughly in preparation for the next time you use the unit.
3. Drain the fuel tank.
4. Start the engine, and leave it in idling condition until all the fuel inside the carburetor is spent and the engine comes to a natural stop.
5. Remove the spark plug and put a few drops (1 to 2 cc) of 2-cycle oil inside the engine. After pulling the starter rope 2 or 3 times, return the spark plugs to their original position and stop in compressed position.
6. After oiling the throttle lever and other metal parts with anti-corrosive oil, cover the sprayer and store indoors in a low-humidity location.

DISPOSAL

When disposing your machine, fuel or oil for the machine, be sure to allow your local regulations.

TROUBLESHOOTING GUIDE

Case 1. Starting failure

CHECK	PROBABLE CAUSES	ACTION
fuel tank	∅ incorrect fuel	∅ drain it and use correct fuel
fuel filter	∅ fuel filter is clogged	∅ clean
carburetor adjustment screw	∅ out of normal range	∅ adjust to normal range
sparking (no spark)	∅ spark plug is fouled/wet	∅ clean/dry
	∅ plug gap is incorrect	∅ correct (GAP: 0.6 to 0.7 mm)
spark plug	∅ disconnected	∅ retighten

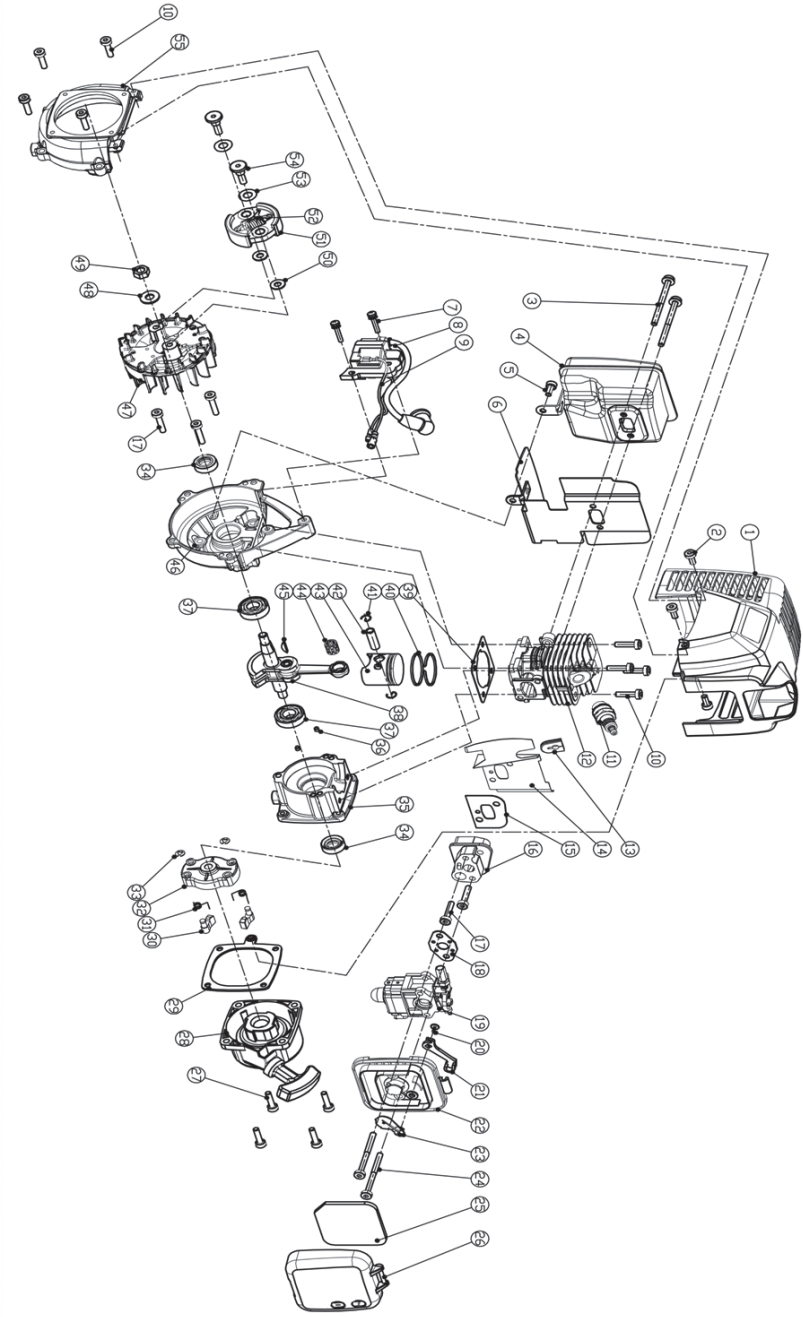
Case 2. Engine starts but does not keep running/hard re-starting

CHECK	PROBABLE CAUSES	ACTION
fuel tank	∅ incorrect fuel or staled fuel	∅ drain it and use correct fuel
carburetor adjustment screw	∅ out of normal range	∅ adjust to normal range
muffler, cylinder (exhaust port)	∅ carbon is built-up	∅ wipe away
air cleaner	∅ clogged with dust	∅ wash
cylinder fin, fan cover	∅ clogged with dust	∅ clean

When your unit seems to need further service, please consult with our service shop in your area.

AGRIMATE® Maximum Power - Super Economy

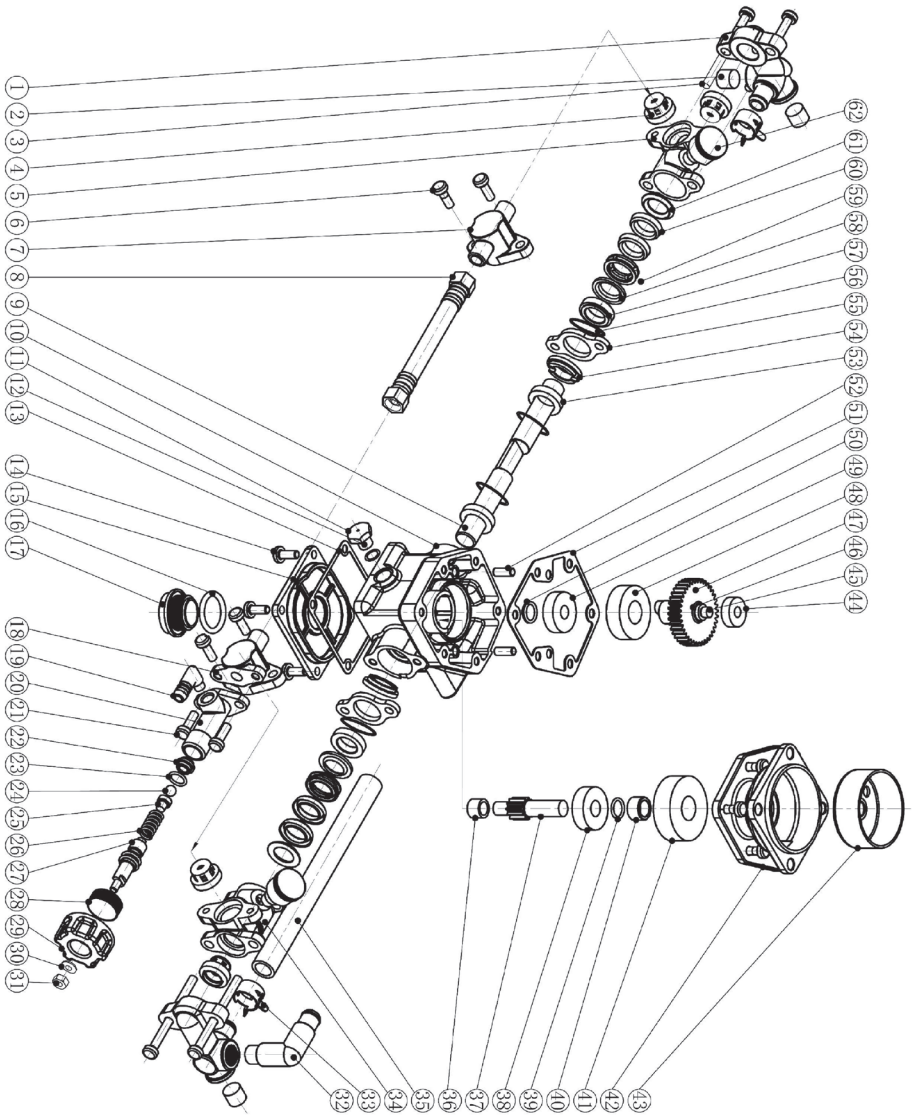
KNAPSACK MOTORISED SPRAYER AM262 ENGINE DIAGRAM



Knapsack Sprayer Engine Part List

Sl. No.	Part No.	Part Name	Sl. No.	Part No.	Part Name
1	AM262-1	Cylinder Cover	29	AM262-29	Sarter Plate
2	AM262-2	Screw	30	AM262-30	Drive Plate Rachet
3	AM262-3	Screw	31	AM262-31	Drive Plate Spring
4	AM262-4	Muffler	32	AM262-32	Drive Plate
5	AM262-5	Screw	33	AM262-33	Open Washer
6	AM262-6	Muffler Gasket	34	AM262-34	Oil Seal
7	AM262-7	bolt	35	AM262-35	Crankcase Starter Side
8	AM262-8	Ignition Coil	36	AM262-36	Location Pin
9	AM262-9	Switch Wire	37	AM262-37	Bearing
10	AM262-10	Screw	38	AM262-38	Crank Shaft assembly
11	AM262-11	Spark Plug	39	AM262-39	Cylinder gasket
12	AM262-12	Cylinder	40	AM262-40	piston ring
13	AM262-13	Rubber pad	41	AM262-41	ring
14	AM262-14	Air Baffle	42	AM262-42	piston pin
15	AM262-15	Intake gasket	43	AM262-43	piston
16	AM262-16	Intake Piple	44	AM262-44	needle case
17	AM262-17	Screw	45	AM262-45	woodruff key
18	AM262-18	Carburater Gasket	46	AM262-46	Crankcase fly wheel side
19	AM262-19	Carburater	47	AM262-47	fly Wheel
20	AM262-20	Location Pin	48	AM262-48	Washer
21	AM262-21	Air clear lock	49	AM262-49	hex nut
22	AM262-22	Air Clear Body	50	AM262-50	plain washer
23	AM262-23	Air clear Baffle	51	AM262-51	clutch assembly
24	AM262-24	Screw	52	AM262-52	spring
25	AM262-25	Filter Cotton	53	AM262-53	Washer
26	AM262-26	Air Clear Baffler	54	AM262-54	clutch bolt
27	AM262-27	Screw	55	AM262-55	side cover
28	AM262-28	Starter Assembly			

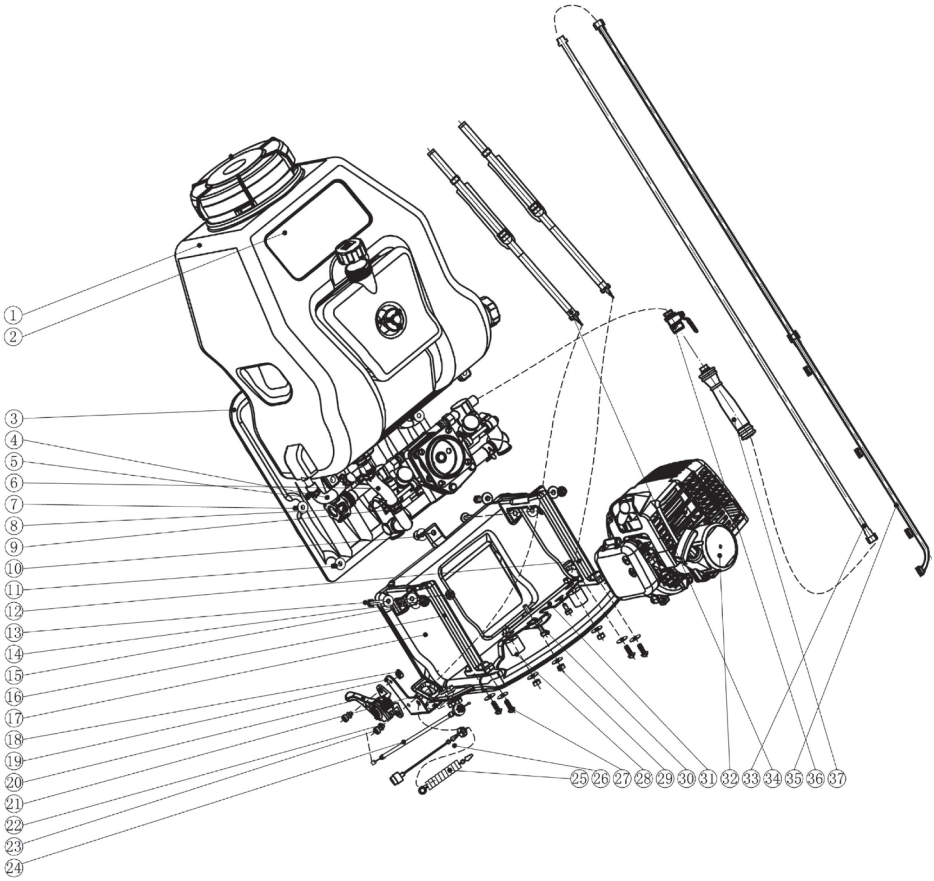
KNAPSACK MOTORISED SPRAYER PLUNGER PUMP ASSEMBLY



KNAPSACK SPRAYER PUMP PART LIST

Sl. No.	Part No.	Part Name
1	AM708TP-1	Connector for inlet
2	AM708TP-2	Plug
3	AM708TP-3	Screw
4	AM708TP-4	One-way valve
5	AM708TP-5	Left body valve
6	AM708TP-6	Screw
7	AM708TP-7	Connector
8	AM708TP-8	High pressure hose
9	AM708TP-9	Plunger
10	AM708TP-10	Pump body
11	AM708TP-11	Plug
12	AM708TP-12	Duckbill valve
13	AM708TP-13	Cushion
14	AM708TP-14	Screw
15	AM708TP-15	Watch window
16	AM708TP-16	O-ring
17	AM708TP-17	Nut of watch window
18	AM708TP-18	Connector
19	AM708TP-19	Nozzle
20	AM708TP-20	Pressure adjusting valve
21	AM708TP-21	Screw
22	AM708TP-22	Spacer
23	AM708TP-23	O-ring
24	AM708TP-24	Steel ball
25	AM708TP-25	Steel ball seat
26	AM708TP-26	Spring
27	AM708TP-27	Pressure adjusting screw
28	AM708TP-28	Pressure adjusting Nut
29	AM708TP-29	Pressure adjusting Handle
30	AM708TP-30	Washer
31	AM708TP-31	Nut
32	AM708TP-32	Elbow
33	AM708TP-33	pipe clip D18
34	AM708TP-34	Right body of pump
35	AM708TP-35	Input pipe
36	AM708TP-36	Needle cage
37	AM708TP-37	Gear shaft 11T
38	AM708TP-38	Bearing
39	AM708TP-39	O-ring
40	AM708TP-40	Liner
41	AM708TP-41	OIL Seal
42	AM708TP-42	Upper cover of Pump body
43	AM708TP-43	Clutch Case
44	AM708TP-44	Bearing
45	AM708TP-45	Gear shaft
46	AM708TP-46	Key
47	AM708TP-47	Gear 36T
48	AM708TP-48	Bearing
49	AM708TP-49	Bearing
50	AM708TP-50	Washer
51	AM708TP-51	Gasket
52	AM708TP-52	Pin
53	AM708TP-53	Oil seal
54	AM708TP-54	Spacer
55	AM708TP-55	Gasket
56	AM708TP-56	O-ring
57	AM708TP-57	Copper spacer
58	AM708TP-58	Rubber washer
59	AM708TP-59	Oil seal
60	AM708TP-60	Y-shaped oil seal
61	AM708TP-61	Washer
62	AM708TP-62	Grease cup

KNAPSACK MOTORISED SPRAYER TANK DIAGRAM



KNAPSACK MOTORISED SPRAYER TANK PART LIST

Sl. No.	Part No.	Part Name
1	AM 708TT-1	Tamk
2	AM 708TT-2	Label
3	AM 708TT-3	Cushior
4	AM 708TT-4	Returhose
5	AM 708TT-5	Clip
6	AM 708TT-6	Inlet hose
7	AM 708TT-7	Screw
8	AM 708TT-8	Plunger pump
9	AM 708TT-9	Clip
10	AM 708TT-10	Bolt
11	AM 708TT-11	Washer
12	AM 708TT-12	right tank supportor
13	AM 708TT-13	Bolt
14	AM 708TT-14	Bolt
15	AM 708TT-15	Nut
16	AM 708TT-16	Left tank supporter
17	AM 708TT-17	Engine seat
18	AM 708TT-18	Nut
19	AM 708TT-19	ON-OFF Swifch seat
20	AM 708TT-20	Throhle lever
21	AM 708TT-21	Nut
22	AM 708TT-22	Washer
23	AM 708TT-23	Bolt
24	AM 708TT-24	Throttle cable
25	AM 708TT-25	Cable
26	AM 708TT-26	On-off switch
27	AM 708TT-27	Screw
28	AM 708TT-28	Bolt
29	AM 708TT-29	Nut
30	AM 708TT-30	Bolt
31	AM 708TT-31	Connetor
32	AM 708TT-32	Eugine
33	AM 708TT-33	Hose
34	AM 708TT-34	Belt
35	AM 708TT-35	Sprayer gun
36	AM 708TT-36	Sprayer switch
37	AM 708TT-37	Sprayer gun grip

AGRIMATE Maximum Power - Super Economy

INSPECTION & MAINTENANCE

For safety use, make sure the machine is cleaned every three-months. Before working, make sure to inspect the filter and the nozzle,and ensure there is no impurity. After working, make sure there is no water in the pump and the hose.

If the machine needs to be repaired in the guarantee period, please return the machine to the shop. You need to provide the certificate of purchase in order to repair or get the substitute.

Before Packing, the machine should be cleaned, then use closed packaging, packaging should be firm. Store the machine in dry place.

For transportation safety. The machine should comply with ISTA.



Meaning of crossed - out wheeled dustbin:

Do not dispose of electrical machines as unsorted municipal waste, use separate collection facilities.

If electrical machines are disposed of in landfills or dumps, hazardous substances can leak into the groundwater and get into the food chain, damaging your health and well-being.

When replacing old machines with new ones, the retailer is legally obligated to take back your old machine for disposals at least free of charge.

Warranty Card

CUSTOMER COPY

Date _____

Purchaser's Name & Address

_____

_____

_____

Product Name

AM-708T Knapsack Motorised Sprayer

Product Serial No.

Invoice / DN No.

Date of Delivery

Dealer's Office Seal

Dealer Sign _____

Customer Sign. _____

I/We accept the terms and conditions of Warranty as in the Warranty Card.

Warranty Card

DEALER'S COPY

Date _____

Purchaser's Name & Address

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Warranty Card

COMPANY COPY

Date _____

Purchaser's Name & Address

_____

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1. LIMITED WARRANTY PERIOD : For a period of six months from the date of Sale / delivery of the product which ever is earlier. 2. LIMITED WARRANTY OBLIGATION : Replacement of defective part(s) by authorized dealer are free of charges. If found defective by reasons of defective material / poor workmanship, parts will be replaced and there will be no value re-imbursement. 3. TERMS & CONDITIONS : a. Claims under warranty will be accepted on submission of Warranty Card duly filled and stamped by authorized dealer together with the original purchase document. : b. Shipping / transportation / Incidental charges incurred in replacing defective parts under warranty shall be borne by the purchaser.

WARRANTY null & void if

(i) Lack of maintenance, (ii) Incorrect use of the machine or tampering, (iii) Use of non-genuine spare parts, (iv) Usage of acids, solvents, or any flammable materials, (v) Non-usage of filter water / clear water / soft water & using water above 40°C temperature, (vi) While using the pressure washers to draw water which is contaminated with dust & solvents (eg. paint thinners, gasoline, oil, etc.), (vii) Repair done by unauthorized service people, (viii) Those parts subject to wear & tear due to normal operations and plastic parts, rubber items etc, (ix) Loose connections, faulty electrical fixtures & fluctuations in power supply.

4. Should any failure occur during or after the warranty period, customer does not have the right to interrupt payment / price discount.
5. Incidental or consequential damages like inconvenience commercial loss, loss of tie, mental agony etc.
6. Loss of any nature due to operation or non-operation is not covered under warranty.
7. All free service are at Dealer's service point only.
8. Warranty ceases if service is not done periodically.
9. The attached card should be completed and returned to Ratnagiri Impex Pvt. Ltd. for registration immediately after sale / delivery.
10. The warranty is void unless you register the warranty card with Ratnagiri Impex Pvt. Ltd.
11. The decision of Ratnagiri Impex Pvt. Ltd., in settling the warranty is final in all respects.
12. All disputes are subject to Bangalore Courts Jurisdiction

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1st Free Service Coupon

ISO 9001-2008
CERTIFIED COMPANY

FREE SERVICE COUPON (30 Days / 20 hrs (Which ever is earlier)

Model: AM-708T Knapsack Motorised Sprayer

Serial Number:

PDI Date | Service No. | Purchase bill no. | Date

Dealer's Office Seal

Operations:
- *Check and Test the Fuel Mixture.*
- *Check and Clean the Air Filter*
- *Check, Clean and Adjust the Spark plug*
- *Check and Adjust the Carburetor Tuning*
- *Check and Sharpen the cutting Tools*
- *Start and Check the Noise, Vibration, Loosen / Fasteners etc.*
- *Product familiarization on Usage*
- *Routine maintenance and Storage*

Dealer Signature.....

Customer Signature.....

Conditions: Operations listed have been performed to my entire satisfaction without any labour charges. Only consumables were charged.

WARRANTY AND FREE SERVICE COUPON

agriculture | horticulture | sericulture | plantations | forestry | garden | health care

2nd Free Service Coupon

ISO 9001-2008
CERTIFIED COMPANY

FREE SERVICE COUPON 60 Days/100 hrs (Which ever is earlier)

Model: AM-708T Knapsack Motorised Sprayer

Serial Number:

PDI Date | Service No. | Purchase bill no. | Date

Dealer's Office Seal

Operations:
- *Check and Test the Fuel Mixture.*
- *Check and Clean the Air Filter*
- *Check, Clean and Adjust the Spark plug*
- *Check and Adjust the Carburetor Tuning*
- *Check and Sharpen the cutting Tools*
- *Start and Check the Noise, Vibration, Loosen / Fasteners etc.*
- *Product familiarization on Usage*
- *Routine maintenance and Storage*

Dealer Signature.....

Customer Signature.....

Conditions: Operations listed have been performed to my entire satisfaction without any labour charges. Only consumables were charged.

WARRANTY AND FREE SERVICE COUPON

agriculture | horticulture | sericulture | plantations | forestry | garden | health care

3rd Free Service Coupon

ISO 9001-2008
CERTIFIED COMPANY

FREE SERVCIE COUPON 90 Days/200 hrs (Which ever is earlier)

Model: AM-708T Knapsack Motorised Sprayer

Serial Number:

PDI Date | Service No. | Purchase bill no. | Date

Dealer's Office Seal

Operations:
- *Check and Test the Fuel Mixture.*
- *Check and Clean the Air Filter*
- *Check, Clean and Adjust the Spark plug*
- *Check and Adjust the Carburetor Tuning*
- *Check and Sharpen the cutting Tools*
- *Start and Check the Noise, Vibration, Loosen / Fasteners etc.*
- *Product familiarization on Usage*
- *Routine maintenance and Storage*

Dealer Signature.....

Customer Signature.....

Conditions: Operations listed have been performed to my entire satisfaction without any labour charges. Only consumables were charged.

WARRANTY AND FREE SERVICE COUPON

agriculture | horticulture | sericulture | plantations | forestry | garden | health care


Small Maintenance. Big Savings.


Small Maintenance. Big Savings.


Small Maintenance. Big Savings.

Do's & Don'ts

- Always read the instruction manual supplied along with the equipments and understand the correct installation and operating procedures before attempting to use the equipments.

- It is important to use the personal protection equipments such as head helmet with visor, ear defenders, goggles, safety gloves, cut resistant shoes, trousers and jackets while operating potentially dangerous machines.

- Avoid untrained persons using the equipments. Children must not be allowed to handle these equipments.

- Always use genuine spares and accessories in repairs and usage to derive maximum life of the equipments.

- Always keep the equipments in good working condition to minimise loss of output as well as to reduce the pollution and save the environment.

- Do not use the machine within range of persons unless they wear protective clothing.

- Do not operate the machine when fatigued or under the influence of alcohol or drugs.

- When the machine is running, please maintain a constant water supply. Without water circulation will damage the sealing rings of the machine.

- For more query kindly contact for our Service Center/Dealer/ franchise

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ANY WHERE - ANY TIME - ANY CROP



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Maximum **Power** - Super **Economy**

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